Daniel Scheidemantel Camargo

Senior Python Backend Engineer | · LLMs · RAG · FastAPI · GCP/AWS

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Brazilian | Married to EU citizen | ***Available immediately***



SUMMARY

Senior Python Backend Engineer with 10 years in software engineering, 8+ of them building and operating high-traffic APIs and data-intensive services on AWS and GCP. Hands-on experience shipping production AI systems: multi-agent architectures, RAG pipelines with hybrid search and reranking, LLM integration (OpenAI, Gemini, Anthropic), embeddings, pgvector and tool-calling workflows. Founded Capellaris AI, an LLM-agnostic multi-tenant SaaS platform for SMBs, as sole engineer alongside a business partner, and won 1st place at the Ignition AI Hackathon (EPFL) with Heatsink, a multi-agent datacenter energy manager. Also brings platform-scale backend experience from Mercado Livre and 4 years as sole technical owner of the backend of a Brazilian travel search platform (4th-largest by web traffic). Strong applied-research background: 13 peer-reviewed publications and an MSc in Applied Computing.

CORE SKILLS

AI / LLM Engineering: LLM integration (OpenAI, Google Gemini, Anthropic Claude); prompt engineering and versioning; tool-calling and function-calling; structured outputs (Pydantic AI); multi-agent orchestration (CrewAI, MCP); agent memory and context management; evaluation and observability

RAG & Vector Search: embeddings pipelines; pgvector; hybrid search (dense + sparse); chunking strategies; reranking; source-attributed answers; multi-tenant knowledge isolation

Back-end & APIs: Python (FastAPI, Django, Flask); Node.js, TypeScript; React, Next.js; C/C++; Java (academic, MSc); Go (limited exposure); REST API design; async services; microservices; serverless functions

Scraping & Automation: Playwright, Selenium, Scrapy, Beautiful Soup; RPA bots and monitoring

Databases & Data: PostgreSQL (+ pgvector), MySQL, Microsoft SQL Server (DBA), BigQuery, Redis, DynamoDB; query optimisation; schema design; ETL pipelines; Polars, Pandas, Dataiku; Plotly Dash, Looker Studio

Cloud & Infrastructure: AWS (EC2, Lambda, Step Functions, RDS, S3, IAM); GCP (Cloud Run, Cloud Functions, App Engine, Firebase, BigQuery); Azure (serverless functions); Heroku; Docker; Kubernetes and Rancher (operated); Linux

CI/CD & Observability: GitHub Actions, Jenkins; Datadog, Grafana; structured logging (Cloud Logging, CloudWatch); prompt versioning

AI coding tools (daily use): Cursor, Claude Code, Codex, Antigravity, OpenCode

Languages: Portuguese – native; English – C1; Spanish – B1; French – A2–B1.

PROJECTS & AWARDS

Heatsink — 1st place, Ignition AI Hackathon (EPFL, Lausanne) **05/2026**
25 teams, 36h, live demos to investors; prize: cash + 1-week residency at the Swiss Data Science Center (SDSC)

- Co-built a simulated 256-rack datacenter where three specialist LLM agents (thermal, workload, energy; Pydantic AI + OpenAI) propose actions in sequence and a deterministic resolver enforces SLA and safety rules.
- Owned datacenter requirements, PUE/cost metrics and simulator realism, plus the manager-facing view (Phaser, TypeScript); **39% lower energy vs. a no-agent baseline in a simulated 10-min failure scenario**.

PROFESSIONAL EXPERIENCE

Senior Software Engineer – Platform & AI Integrations
Mercado Livre *(logistics network & facility simulators, ∼5k internal users, LATAM)* **08/2025 – 08/2026**

- Integrated OpenAI and Claude APIs as **decision-support agents** in a production simulation platform used by ∼5k supply-chain specialists across LATAM: agents validate user-entered parameters and suggest corrections (∼**25% fewer input errors** vs. before the agents) and interpret computed KPIs by comparing a summary of the full dataset against the desired behaviour.
- Evaluated the agents on **p95 latency under 30s, user-rated quality (4.6/5 average)** and token consumption per user; own the lifecycle from prompt design and versioning through deployment and iteration.
- Redesigned the simulation data-input flow, splitting one large spreadsheet into smaller sequential steps, **reducing input errors by 17%**.

- Built data pipelines over **terabytes in BigQuery**: chained queries and transformations that break down real historical demand and apply leadership growth forecasts by category, region and facility type, producing demand projections over an adjusted logistics network.
- Designed the **async FastAPI** layer between frontend and backend: receives user parameters, handles database operations and job status without blocking other requests, so the application keeps scaling.
- Moved data processing from Pandas to **Polars**, absorbing a **2x increase in input rows** from user-submitted Excel files.
- Used **Dataiku** for transformations holding business rules owned by domain specialists, so rules could be edited without a code deploy.
- Mentored 2 mid-level engineers in an 8-person team; one was promoted to senior.
- Maintained CI/CD pipelines (**GitHub Actions**) and observability (**Datadog**, **Grafana**); added new metrics through new endpoints.
- Stack: Python, FastAPI, BigQuery, MySQL, Dataiku, Docker, React (front-end); limited exposure to Go (edited one microservice).
- Placed **4th of 22 teams** in an internal AI hackathon (02/2026, team of 6), using **n8n** to evaluate operational metrics.

Founder & Lead AI/Backend Engineer
Capellaris AI *(SaaS, agentic AI platform for SMBs)* **11/2024 – 09/2025**

- Designed and built an **LLM-agnostic multi-tenant agentic SaaS platform** end-to-end as sole engineer (business partner owned commercial, financial and business processes), enabling SMBs to deploy specialised AI agents across business domains; also owned post-sales and customer training.
- Built a full **RAG pipeline**: document ingestion, chunking, embedding generation, pgvector storage, hybrid search, reranking and source-grounded answers, reused across all agent domains.
- Implemented a **hierarchical multi-agent orchestration layer** (manager agent + specialists; CrewAI + MCP) with tool-calling and memory management.
- Designed a **two-tier agent memory**: long-term RAG over business rules, the business plan and previously well-rated answer formats (appended to the vector DB), plus dynamic short-term memory where an automatic cyclic summary of the chat and prior context is written at **60% of the agent's context window**, prioritising newer context.
- Tracked a **1–5 star user rating** for every project delivered by the manager agent, from the first single-agent deliveries through hierarchical multi-agent runs: later projects scored higher and needed **fewer refinement rounds**.
- Piloted with **three small businesses** (an independent consultancy, a restaurant and an art atelier) plus internal use.
- Delivered financial control agents that gave a pilot customer a **unified view of financial planning** across departments, replacing fragmented manual processes.
- Implemented the MLOps lifecycle: model integration, prompt versioning, evaluation, observability and **CI/CD with GitHub Actions**.
- Full-stack delivery: **React** front-end with a **Node.js** backend-for-frontend (BFF).
- Infrastructure: GCP (Cloud Run, Firebase), Supabase PostgreSQL + pgvector, Cloudflare; OAuth 2.0, IAM, multi-tenant secrets management.

Senior Python Backend / Platform Engineer
Buscaônibus / Travelier *(travel meta-search and BI platform)* **04/2021 – 01/2025**

- Sole technical owner of a high-traffic Python backend processing **millions of daily searches** across hundreds of bus operators (on-call, incident response, full production lifecycle); platform ranked 4th-largest in Brazil by web traffic (SEMrush).
- Led a complete platform migration (**Python 2.7 to 3.10, Django 1.9 to 5, App Engine to Cloud Run**) in well-defined stages, using **feature flags backed by cached keys** for immediate rollback; new complex APIs on Cloud Run were built in **FastAPI** and specific microservices as **Flask** Cloud Functions. **Average API latency −25%** and **cloud cost −40%** (mean before vs. after the migration).
- Redesigned the **Redis caching layer**: all database and partner data cached with per-source TTLs, structured key management, and a Redis manager to inspect and manipulate keys. Reached a **92% cache hit ratio** (GCP Redis metric, 30-day average).
- Built on-demand **scrapers** (Python, Playwright, Selenium, Beautiful Soup on Cloud Run/Docker) for 5 small bus companies without digital integration, and built a monitoring dashboard that surfaced when a company stopped returning data.
- Built a **Gemini RAG agent** with engineered prompting to automate triage and responses for customer emails from the website contact forms, with responses rated **89% accurate on average** by the support specialist who handled these emails.
- Designed a BigQuery analytics layer delivered through **Plotly Dash** dashboards (Pandas/NumPy) and Looker Studio views: market, competitor and route-coverage intelligence for partner bus companies' strategic decisions, plus an internal sales BI for the commercial team.
- Mentored a junior engineer (a content copywriter turned frontend developer) toward full-stack on seniority, project management, ticket lifecycle and pull-request practices.

Software Engineer – Integrations & Backend
Portobello Group *(ceramics manufacturing & retail, ∼4,000 employees)* **12/2019 – 04/2021**

- Built and maintained a **regional fiscal discount simulator** (Python/Django on **Heroku** with **PostgreSQL**) that calculates the full tax burden of lists of complex, diverse products, integrated into Salesforce (Apex API layer and UI button) and rendered there as HTML served by the service; worked with fiscal stakeholders to raise **fiscal-rule coverage from 70% to 96%**.
- Part of the team that delivered the company's 2021 digital transformation, including the CRM/ERP migration to **Salesforce** across **124 retail stores (mostly franchises)**, with **no integration-related incident tickets** (only usage questions); designed the product data model and the product-to-order connection.
- Administered the **Microsoft SQL Server** database that held company-wide data and integrated multiple ERPs and CRMs (ETLs, transformations) ahead of the Salesforce migration, hosted on an **AWS EC2** virtual machine.
- Worked across **Microsoft Azure**, where every new service was a microservice (serverless functions and a data pipeline), and **AWS** (SQL Server on EC2).
- Operated **Kubernetes (Rancher)** and **Jenkins** in an API-management integration epic with attached microservices.

Software Developer – Automation & Cloud
Stormtech *(consumer loans, regulated)* **08/2018 – 11/2019**

- Developed RPA bots (Python, Scrapy, Selenium, Beautiful Soup) that scrape Brazilian loan-bank portals to update loan status automatically in an internal PHP loan-management system; wrote about half of the bots and maintained the rest, updating them whenever a bank changed its contract or portal.
- Implemented a **TypeScript dashboard** with real-time visibility of bot health (running, failed, unused), cutting the time to identify a broken pipeline from about 2 hours to under 1 hour. The dashboard also showed stability per bank and alerts.
- Automated roughly half of the operators' manual workload by adding bots for new banks.
- Ran the bots mostly on **AWS Lambda** (EC2 for those requiring Windows), with **Step Functions**-orchestrated Lambda ETL feeding **DynamoDB** (bank update data) and MySQL (client data and security layer), across 18 integrated banks.

Internet of Things Engineer
Comigo LTDA *(startup, cloud-connected reverse vending machine, beta stage)* **10/2016 – 06/2018**

- Developed the edge software of a beta **reverse vending machine** for PET bottles and aluminum cans: a conveyor brings each item to a camera where **Python + OpenCV** reads the barcode, shape and label; after validating product and user against the database, the machine credits points to the user.
- Wrote the **Arduino (C++)** firmware for motors and presence sensing and the **Raspberry Pi** Python and Shell scripts (GPIO automation) that tie the microcontroller, an **RFID** reader/keypad and the cloud together.
- Designed and configured the company's **AWS** environment (EC2 on Linux, RDS MySQL, S3, security groups, IAM roles) accessed over Wi-Fi/HTTP by the machine; managed the developer who designed the database schema and built the HTML/JavaScript/CSS front-end.
- Took part in startup and product definition applying Lean Startup practices (MVP, Design Thinking, business model canvas, competitor analysis, business plan led by the CEO). Team of five; single unit tested in a controlled environment; the company closed before public launch.
- Trello for task management and Bitbucket for version control.

EDUCATION

Master of Science in Applied Computing **08/2016 – 07/2018**
Santa Catarina State University (UDESC), Brazil – Thesis: *Resource Consolidation with High Availability on Virtualized Environments*. Distributed Systems, Cloud Computing, Java, data-centre efficiency (6 peer-reviewed articles – full list of 13 publications: Google Scholar).

Bachelor in Computer Science **02/2011 – 07/2016**
Santa Catarina State University (UDESC), Brazil – IoT, automation, energy efficiency (7 peer-reviewed articles)

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